DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION
Year 6

HIGHER 1 GEOGRAPHY **8834/01**

INDICATIVE CONTENT

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1 (a) Suggest a suitable research question for the investigation and explain why it is suitable. [4]

A suitable research question is: *"Is the initiative 'Reuse and Return Project' in the Tampines Hawker Centre effective in encouraging the use of reusable containers?" [**1 mark**]*

It is suitable because the research question is:
- clearly defined [**1 mark**]
- capable of being researched [**1 mark**]
- of a suitable scale [**1 mark**]

(b) Explain some possible issues related to risks and research ethics about the students' data collection shown in Resource 1. [5]

Ethical Concerns
- Participants and the hawker may be unaware of the video recording taking place [**1 mark**]
- Consent may not have been obtained from participants [**1 mark**]
- Data confidentiality may be breached [**1 mark**]

Risks
- Students may be injured when customers walk with their food products [**1 mark**]
- Loss or damage of equipment [**1 mark**]

(c) With reference to Resource 2, describe the results for the customers' choice of container for food to go. [5]

- In general, only a small proportion of customers used reusable container [**1 mark**]
- Customers who used Reusable Container
  - Most customers who used reusables did so because they cared for environment and being cost conscious [**1 mark**].
- Customers who used Single Use Container
  - Almost all customers who used single-use container disagreed that they did not care for the environment [**1 mark**].
  - All customers using one-off packaging felt that convenience is important [**1 mark**]
  - Most customers disagreed/strongly disagreed that cost is important. [**1 mark**]

(d) The students concluded that the initiative has not been effective.

With reference to all resources, evaluate the validity of their conclusion. [10]

Candidates could refer to the Resources and consider how accurate and reliable the collected data are for drawing conclusion. Candidates could look at the way data are collected and assess if the methods used could make accuracy and reliability of the data questionable. For instance, they could look at whether the collection of data over one day at one timing and in only one part of the year is sufficient in providing robust data for their investigation. Information from Resources and preamble could be used to infer the factors that could contribute to making the data in other Resources inaccurate and unreliable.

Additionally, candidates could consider what other data would be needed for this investigation. For instance, they could mention the need for information on the respondents' dining habits in view that this is a key element in the choice of takeaway.

Levels marked using Generic Level Descriptors for H1 fieldwork evaluative Questions.

(e) The town council has asked the group to gather information that will help to reduce single-use plastic packaging in supermarkets in a neighbourhood. Explain how the group would go about gathering the information. [6]

Students should establish the primary and secondary data needed to examine the scenario posed. They could outline the secondary data such as the number and types of supermarkets present in the neighbourhood so as to know how to properly sample [**1 mark**].

To ensure validity of the questionnaire, the instrument should include both closed & open-ended questions [**1 mark**]. Possible open-ended questions would be '*What would encourage you to use less plastic packaging in supermarket?*' would allow us to find out the main reason behind the customer's decision [**1 mark**].

Sampling method should be established (e.g. stratified sampling). This could be based on strata such as income level [**1 mark**].

Students should indicate how the questionnaire can be administered. They can conduct the survey over a 2-hour period and over both weekdays and weekends over a month [**1 mark**].

The students could outline the risks and/or ethics of undertaking this research [**1 mark**].

2 **(a)** Compare the urban land-use models shown in Resources 4A and 4B. [5]

- Both the CBD are in the centre of the urban area/city [**1 mark**]
- Both DC and LDC have high class housing, slum housing, CBD and industry land use [**1 mark**]
- DC only has rings but LDC has both rings and wedges [**1 mark**]
- Only DC has mid-class housing while LDC does not have it [**1 mark**]
- No recent slums in DC model but there are new slums in LDCs [**1 mark**]

Accept all possible answers

(b) Suggest possible reasons for the locations of the slum housing seen in Resources 4A and 4B. [4]

There is a shift from traditional heavy industry [**1 mark**]. To cut down on cost for transportation [**1 mark**]
The wealthy residents shifted away [**1 mark**].
Rapid growth in urban population [**1 mark**]

(c) Suggest reasons why developed countries such as England undertake urban reimaging efforts. [4]

Post-industrial cities typically went through a phase of deindustrialization where traditional manufacturing industries declined [**1 mark**]. To arrest the decline, reimaging was embarked to attract investment back [**1 additional mark**]

To attract capital [**1 mark**]. To create an image of a desirable place [**1 additional mark**]

(d) Using Resources 5A and 5B, suggest ways in which the changes shown between 1979 and 2016 have improved sustainable development of Newcastle-upon-Tyne. [5]

- Pedestrianisation, no buses, cars, vans, parking spaces [**1 mark**]

- Change of Pavement widening of surface allows for safer environment for those who are disabled [**1 mark**]
- Buskers add ambiance and enhance the overall environmental quality [**1 mark**]
- A broader economic base of the city to allow for broader range of jobs [**1 mark**]
- Cleaner and more pleasant environment [**1 mark**]

Accept any suitable answers

(e) Explain the concept of sustainable development, as defined in 'Our Common Future'. [4]

Meeting of present needs
- Many of the poor's basic needs remains unmet. The poor are vulnerable [**1 mark**].
- Sustainable development aims to ensure more inclusivity and ensuring social well-being of the vulnerable population is achieved [**1 additional mark**].

Meeting future needs
- Future generations should inherit a planet with the resources necessary for their well-being [**1 mark**].
- This aims to ensure that future generation has adequate resources for development [**1 mark**].

3 (a) Using Resource 6, describe the changes to the position of the ice front of the Larsen B Ice Shelf, Antarctica, from 1947 to 2002. [5]

- Overall steady retreat between 1947 and 2002 [**1 mark**]
- Retreat was smaller earlier between 1947 and 1961 [**1 mark**]
- Retreat was largest between 1993 and Jan 2002 [**1 mark**]
- Retreat was most rapid between 31 Jan 2002 and 5 March 2002 [**1 mark**]
- However, between 1961 and 1993, there was an advance of ice front [**1 mark**]

(b) Explain how changes in ice sheet and one other natural factor can influence temperatures through feedback mechanisms. [4]

- With climate change, temperature increases → more ice melts, resulting in lower albedo [**1 mark**]. A positive feedback occurs, resulting in further warming [**1 additional mark**].
- With climate change, temperature increases → more ice melts, resulting in lower salinity [**1 mark**]. Weakens thermohaline circulation → Less heat is transported to higher latitudes, resulting in negative feedback [**1 additional mark**].

(c) Using Resource 7, describe the distribution of the number of people living on land predicted to be below sea level by 2100. [5]

- Most regions (except Africa) are at risk of flooding having at least 0.1 million people at risk [**1 mark**]
- Region with greatest number of people found in Asia [**1 mark**]
- Areas with lesser number of people are least common [**1 mark**]
- Areas with even lesser number million people are found in all regions [**1 mark**].
- Areas with lowest risk are predominantly found in coastal and interior Africa [**1 mark**]

(d) Describe the changes in the mean sea surface temperatures compared to the 20th century average shown in Resource 8A. [4]

- Temperature difference increased [**1 mark**]
- Changes between 1880 to 1940: temperature difference were generally negative [**1 mark**]
- Changes between 1950 and 1975: temperature difference fluctuated around 0◦C [**1 mark**]
- Changes between 1975 and 2020: generally positive temperature difference [**1 mark**]

(e) With reference to Resource 8B, suggest possible impacts of climate change on aquatic ecosystems. [4]

- Average temperatures increase **and** lobsters start to migrate northwards away from original location as the seas are cooler at higher latitudes [**1 mark**] This can result in the disruption of ecosystems in both the lower and higher latitudes [**1 additional mark**]

- Eutrophication may become more common. [**1 mark**]. Algae blooms that occur can be expected to increase in frequency, which can reduce water quality [**1 additional mark**]

Accept any suitable content.

4 'States play a more important role than firms in responding to climate change.' Evaluate the statement. [13]

Candidates could approach the question by making a judgement on whether states play a more important role than states through a consideration of the scale of actions/outcomes, immediacy of the actors' actions/outcomes and how long lasting the actions/outcomes are. These actions could be aimed at addressing different responses to climate change such as those related to mitigation and adaptation to impacts of climate change.

5 Evaluate the success of the strategies used to address the issues faced by elderly living in cities.

Candidates could approach the question by making a judgement on whether some strategies are more effective than others through a consideration of the relative strengths and limitations of the strategies. The strategies could be compared in relation to criteria such as costs, benefits and ability to target a wide range of issues. These strategies could be aimed at addressing different issues related to elderly such as those related to economic well-being, social well-being and psychological well-being. They could also include those implemented at different scales or by different stakeholders.

Candidates could also approach the question by making a judgement on whether the same strategies have been more effective in some places than others through a consideration of two or more case studies. Candidates could evaluate whether the strategies in each case achieved the intended aims. Candidates could also analyse the unintended negative impacts of the strategies. Candidates could also analyse the contextual factors such as the scale of the issues, the political will of the authorities and the resources available, which may have contributed to the variations in success between these places.

6 'Strategies to improve the lives of slum dwellers have never been successful in cities from developing countries.' Evaluate the statement.

Candidates could approach the question by making a judgement on whether the strategies have never been successful through a consideration of the relative strengths and limitations of the strategies. The strategies could be compared in relation to criteria such as scale and time. Strategies may also vary also space. These strategies could be aimed at addressing different issues related to slum dwellers.